

Appendix B: Units Used in This Document

This datasheet follows standard practice for use of SI units as recommended by NIST, except in the context of memory and storage capacity. Here it adopts the convention that the prefixes k (*kilo*), M (*mega*) and G (*giga*) always refer to the nearest power of two to their standard decimal value. This aligns the datasheet with common usage for these units.

Memory and Storage Capacity

This datasheet expresses memory and storage capacity using the following units:

- b: bit
 - kb: kilobit, $2^{10} \text{ b} = 1024 \text{ b}$
 - Mb: megabit, $2^{20} \text{ b} = 1,048,576 \text{ b}$
 - Gb: gigabit, $2^{30} \text{ b} = 1,073,741,824 \text{ b}$
- B: byte, eight bits, one octet
 - kB: kilobyte, $2^{10} \text{ B} = 1024 \text{ B}$
 - MB: megabyte, $2^{20} \text{ B} = 1,048,576 \text{ B}$
 - GB: gigabyte, $2^{30} \text{ B} = 1,073,741,824 \text{ B}$

The bit is the most basic unit of information. A bit is either true (1) or false (0).

Transfer Rate

Units for transfer rate are dimensionally the product of one byte or bit with a unit of frequency such as MHz. Therefore the standard SI prefixes apply:

- b/s: bit per second
 - kb/s: kilobit per second, 1000 b/s
 - Mb/s: megabit per second, 1,000,000 b/s
 - Gb/s: gigabit per second, 1,000,000,000 b/s
- B/s: byte per second, 8 b/s
 - kB/s: kilobyte per second, 1000 B/s
 - MB/s: megabyte per second, 1,000,000 B/s
 - GB/s: gigabyte per second, 1,000,000,000 B/s

Physical Quantities

The following units express physical quantities such as voltage and frequency:

- A: ampere, unit of electrical current, one coulomb per second

- mA: milliampere, 10^{-3} A
 - μ A: microampere, 10^{-6} A
- Ω : ohm, unit of electrical impedance or resistance, one volt per ampere
 - M Ω : megohm, 10^6 Ω
 - k Ω : kilohm, 10^3 Ω
 - m Ω : milliohm, 10^{-3} Ω
- V: volt, unit of electrical potential difference, one joule per coulomb
 - kV: kilovolt, 10^3 V
 - mV: millivolt, 10^{-3} V
 - μ V: microvolt, 10^{-6} V
- Hz: hertz, unit of frequency, one period per second (s^{-1})
 - kHz: kilohertz, 10^3 Hz
 - MHz: megahertz, 10^6 Hz
 - GHz: gigahertz, 10^9 Hz
- F: farad, unit of electrical capacitance, one coulomb per volt
 - mF: millifarad, 10^{-3} F
 - μ F: microfarad, 10^{-6} F
 - nF: nanofarad, 10^{-9} F
 - pF: picofarad, 10^{-12} F
- H: henry, unit of electrical inductance, one volt-second per ampere (VsA^{-1})
 - mH: millihenry, 10^{-3} H
 - μ H: microhenry, 10^{-6} H
 - nH: nanohenry, 10^{-9} H
- s: second, unit of time
 - ms: millisecond, 10^{-3} s
 - μ s: microsecond, 10^{-6} s
 - ns: nanosecond, 10^{-9} s
 - ps: picosecond, 10^{-12} s
- J: joule, unit of energy or work, one newton-metre
- C: coulomb, unit of electrical charge
 - mC: millicoulomb, 10^{-3} C
 - μ C: microcoulomb, 10^{-6} C
 - nC: nanocoulomb, 10^{-9} C
- m: metre, unit of length or distance
 - mm: millimetre, 10^{-3} m
- $^{\circ}C$: degree celsius, unit of temperature
- W: watt, unit of power, one joule per second (Js^{-1}) or one volt-ampere (VA)

- mW: milliwatt, 10^{-3} W
- μ W: microwatt, 10^{-6} W
- nW: nanowatt, 10^{-9} W

Scale Prefixes

The standard SI prefixes used in the previous sections are:

- G: giga, factor of 10^9 (one short billion)
- M: mega, factor of 10^6 (one million)
- k: kilo, factor of 10^3 (one thousand)
- c: centi, factor of 10^{-2} (one hundredth)
- m: milli, factor of 10^{-3} (one thousandth)
- μ , micro, factor of 10^{-6} (one millionth)
- n: nano, factor of 10^{-9} (one short billionth)
- p: pico, factor of 10^{-12} (one short trillionth)

The customary binary prefixes used in the memory and storage capacity section are:

- G: giga, factor of 2^{30} (approximately 10^9)
- M: mega, factor of 2^{20} (approximately 10^6)
- k: kilo, factor of 2^{10} (approximately 10^3)

These customary binary prefixes are equivalent to the following prefixes from IEC 60027-2:

- Gi, gibi
- Mi, mebi
- Ki, kibi

Digit Separators

Numbers written out with many digits may have either commas or spaces inserted for easier reading:

- 1,000,000: one million
- 1 000 000: one million

A comma in a number *never* represents a decimal (radix) point.